DTC P1284:00 [SKYACTIV-D 1.5]

id0102q2128400

DTC P1284:00	Fuel injector carbon deposit burn-off procedure not completed
DETECTION CONDITION	• The fuel injector carbon deposit burn-off procedure was not completed normally. **Diagnostic support note** • This is a continuous monitor (CCM). • The check engine light illuminates if the PCM detects the above malfunction condition during the first drive cycle. • FREEZE FRAME DATA/Snapshot data is available. • DTC is stored in the PCM memory.
FAIL-SAFE FUNCTION	• Not applicable
POSSIBLE CAUSE	• Any of the following conditions is met during fuel injector carbon deposit burn-off — Connection between DLC-2 and M-MDS becomes loose — Engine coolant temperature: **70 °C {158 °F} or less** — Ambient temperature (servicing area temperature): **-2 °C {28 °F} or less** — Intake air temperature: **3 °C {37 °F} or less** — Ignition switched OFF — DTCs other than the following DTC are detected • Abnormal diesel particulate filter regeneration interval short distance: P2459:00 • Fuel injector carbon deposit burn-off is not completed because the engine load and the fuel injection amount are low due to any of the following conditions. — Engine compression pressure is low (load by pumping is low) — A/C load is low (in FRESH mode) — A/C malfunction • Fuel injector carbon deposit burn-off procedure not completed or procedure irregularity • Fuel injector malfunction • PCM malfunction

Diagnostic Procedure

STEP	INSPECTION		ACTION
1	**RECORD VEHICLE STATUS WHEN DTC WAS DETECTED TO UTILIZE WITH REPEATABILITY VERIFICATION** **Note** • Recording can be facilitated using the screen capture function of the PC. • Record the freeze frame data/snap shot data.	—	Go to the next step.
2	**VERIFY RELATED SERVICE INFORMATION AVAILABILITY** • Verify related Service Information availability. • Is any related Service Information available?	Yes	Perform repair or diagnosis according to the available Service Information. • If the vehicle is not repaired, go to the next step.
		No	Go to the next step.
3	**VERIFY RELATED PENDING CODE AND/OR DTC** • Switch the ignition off, then ON (engine off). • Perform the Pending Trouble Code Access Procedure and DTC Reading Procedure. (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-D 1.5].) • Are any DTCs present?	Yes	Repair the malfunctioning location according to the applicable DTC troubleshooting, then go to the next step. (See DTC TABLE [SKYACTIV-D 1.5].)
		No	Go to the next step.
4	**INSPECT DLC-2 CONNECTION CONDITION** • Switch the ignition off. • Inspect the connection between the DLC-2 and M-MDS, and inspect the terminals for damage, deformation, corrosion, or disconnection. • Is there any malfunction?	Yes	Repair or replace the malfunctioning part according to the inspection results, then go to the next step.
		No	Go to the next step.

STEP	INSPECTION		ACTION
5	**VERIFY FUEL INJECTOR CARBON DEPOSIT BURN-OFF CONDITION AND PERFORM AGAIN** **Note** • Perform the procedure from Step 18 (fuel injector carbon deposit burn-off start) in the fuel injector carbon deposit burn-off procedure because the fuel injector carbon deposit burn-off was unsuccessful. • Start the engine and warm it up completely. • Verify that all of the following conditions are met. — Engine coolant temperature: **80 °C {176 °F} or more** — Ambient temperature (servicing area temperature): **0 °C {32 °F} or more** — A/C REC • Perform the procedure from Step 18 (fuel injector carbon deposit burn-off start) in the "FUEL INJECTOR CARBON DEPOSIT BURN-OFF PROCEDURE". (See FUEL INJECTOR CARBON DEPOSIT BURN-OFF PROCEDURE [SKYACTIV-D 1.5].) • Was the fuel injector carbon deposit burn-off procedure successful?	Yes	DTC troubleshooting completed.
		No	Go to the next step.
6	**PERFORM PROCEDURE FROM STEP 18 (FUEL INJECTOR CARBON DEPOSIT BURN-OFF START) IN FUEL INJECTOR CARBON DEPOSIT BURN-OFF PROCEDURE** **Note** • It is necessary to perform the procedure from Step 18 (fuel injector carbon deposit burn-off start) in the fuel injector carbon deposit burn-off procedure due to the possibility of low engine load. • Perform the procedure from Step 18 (fuel injector carbon deposit burn-off start) in the "FUEL INJECTOR CARBON DEPOSIT BURN-OFF PROCEDURE". (See FUEL INJECTOR CARBON DEPOSIT BURN-OFF PROCEDURE [SKYACTIV-D 1.5].) • Was the fuel injector carbon deposit burn-off procedure successful?	Yes	Go to the next step.
		No	If it was not completed, replace the fuel injector in all of the cylinders, then go to the next step. (See FUEL INJECTOR REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
7	**VERIFY DTC TROUBLESHOOTING COMPLETED** • Always reconnect all disconnected connectors. • Clear the DTC from the PCM memory using the M-MDS. (See CLEARING DTC [SKYACTIV-D 1.5].)	—	DTC troubleshooting completed.